Near-Null Magnetic Fields in Lunar Plant Biology: The First 5 Days of Growth

1. LUNAR MAGNETIC REALITY

- Moon lacks global core dynamo
- Artemis South Pole: < 0.8 nT
(Extremely near-null vector)
- Btotal = Bcrust + Bext +
Blander + Bpayload + Bcontrol
- Lander avionics (50–500 μT)
confound hypomagnetic flight

Core Concept 1:
Physical Reality of Lunar HMF

2. FIRST 5 DAYS OF GROWTH

- 0–72 h Heterotrophic Reserve:
Beta-ox. + Glyoxylate + Mito
- Linear Seedling Soot (Greens)
- Delayed Emergence (Δt50)
- 25–30% Early Biomass Penalty
- Repressed Pigment Biosynthesis:
HEMA1, CHLH, POR, CHS, HY5

Core Concepts 2, 3, 4:
Viability, Speed, Pigments

3. MULTI-STRESSOR NEXUS

- Exploration Atmosphere (8.2 psia,
34% O_2, elevated CO_2)
- GCR Ionizing Radiation
- Mitochondrial ROS Amp:
AOX1D & CAT3 induction
- Cross-kingdom parallel with
astronaut mitochondrial stress
- Predeclared Flight Gates (G/A/R)

Core Concept 5:
CO2 <-> O2 <-> ROS Nexus